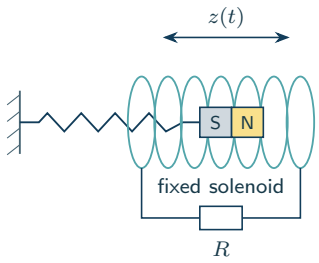
MOTION, INDUCTION AND THE QUADRATIC MEAN

RMS current from an oscillating magnet

Motion changes flux linkage; the mean square of current measures heating.

A magnet on a spring



Motion ⟶ changing linkage ⟶ current

The magnet's flux linkage is $\Lambda_m(z) = \sum_n \Phi_{m,n}(z)$; it equals $N\Phi_m$ if all turns link the same flux.

$$z(t) = z_0 + A\cos\omega t, \qquad f = \frac{\omega}{2\pi},$$

$$\mathcal{E}_m(t) = -\frac{d\Lambda_m}{dt} = -\Lambda_m'(z(t))\,\dot z(t).$$

Close the loop through total resistance R (coil plus load). Neglecting inductance, $i \simeq \mathcal{E}_m/R$ [1]. Use steady driven motion, or one cycle with weak damping.

A. Nearly constant flux gradient

Over a small range about z_0, approximate

$$\Lambda_m(z) \simeq \Lambda_0 + g(z - z_0), \qquad g = \Lambda_m'(z_0).$$

Since $\dot z = -A\omega\sin\omega t$,

$$i(t) \simeq \frac{gA\omega}{R}\sin\omega t, \qquad I_p = \frac{|g|A\omega}{R}.$$

This current is sinusoidal at electrical angular frequency $\Omega = \omega$. Hence

$$\boxed{I_{\rm rms} = \frac{|g|A\omega}{\sqrt{2}\,R}.}$$

B. Centred in a symmetric coil

With $z_0 = 0$, symmetry gives $\Lambda_m'(0) = 0$. The leading position-dependent term is quadratic:

$$\Lambda_m(z) \simeq \Lambda_0 + \tfrac{1}{2}\kappa z^2, \qquad \kappa = \Lambda_m''(0).$$

Therefore, to leading order,

$$i(t) \simeq \frac{\kappa A^2\omega}{2R}\sin 2\omega t, \qquad \boxed{I_{\rm rms} = \frac{|\kappa|A^2\omega}{2\sqrt{2}\,R}.}$$

Here $\Omega = 2\omega$: the electrical frequency is $2f$. SHM alone does not fix the current waveform.

RMS: square, average in time, then take the root

RMS is the **quadratic mean**. With electrical period T_e, the weights are fractions of time:

$$I_{\rm rms}^2 = \frac{1}{T_e}\int_0^{T_e} i^2(t)\,dt \simeq \sum_j w_j i_j^2, \quad w_j = \frac{\Delta t_j}{T_e}, \quad \sum_j w_j = 1.$$

For $i(t) = I_p\sin\Omega t$, the complementary shares $p = \sin^2\Omega t$, $q = \cos^2\Omega t$ obey $p + q = 1$. A quarter-cycle shift swaps them, so their cycle averages are equal:

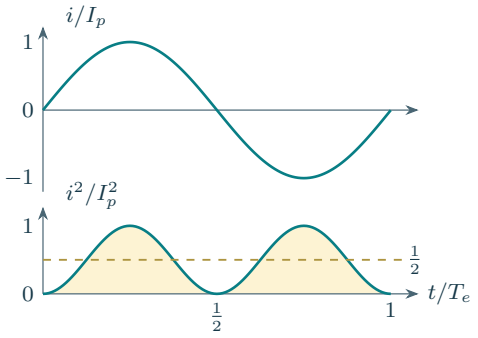
$$\langle p\rangle = \langle q\rangle = \tfrac{1}{2}, \qquad \boxed{I_{\rm rms} = \frac{I_p}{\sqrt{2}}.}$$

The signed mean current is zero. Heating depends on the mean square:

$$\boxed{\overline{P} = R\langle i^2\rangle = RI_{\rm rms}^2.}$$

Thus $I_{\rm rms}$ is the equivalent DC heating current [2].

One electrical cycle



Top: $\langle i\rangle = 0$.
Bottom: $\langle i^2\rangle = I_p^2/2$.

Example: the constant-gradient case

Take $g = 0.50\ \mathrm{Wb\,m^{-1}}$, $A = 0.020\ \mathrm{m}$, $f = 2.0\ \mathrm{Hz}$ and $R = 10\ \Omega$. Then

$$\omega = 2\pi f, \qquad \mathcal{E}_p = |g|A\omega = 0.126\ \mathrm{V},$$

$$I_p = 12.6\ \mathrm{mA}, \qquad \boxed{I_{\rm rms} = 8.9\ \mathrm{mA}.}$$

The average heating is $\overline{P} = 0.79\ \mathrm{mW}$. The gradient g already includes all coil turns.

Use measured samples for a distorted trace

With sufficiently dense, equally spaced samples over whole electrical cycles,

$$\boxed{I_{\rm rms} \simeq \sqrt{\frac{i_1^2 + i_2^2 + \cdots + i_n^2}{n}}.}$$

Measure V_j across a series sense resistor R_s and use $i_j = V_j/R_s$. Moving-magnet coils can produce pulses [3]; the $I_p/\sqrt{2}$ shortcut requires a sinusoidal **current** trace.

With inductance [2]. For a steady sinusoidal emf of peak $\mathcal{E}_p$ and angular frequency Ω, with constant R, L, $I_{\rm rms} = \mathcal{E}_p/\sqrt{2[R^2 + (\Omega L)^2]}$.

Sources. [1] OpenStax, *University Physics 2*, §13.1. [2] §15.4 (RMS/power) and §15.3 (impedance). [3] A. Singh et al., *Electromagnetic induction and damping*, physics/0111016 (2001).